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Aim : Aim: MANET Simulation with Directional Antenna

Simulate MANET using directional antennas and analyze improvements in spatial reuse and interference reduction.

MANET Simulation with Directional Antenna

set ns [new Simulator]

Trace files

set tracefd [open manet.tr w]

\$ns trace-all \$tracefd

set namfd [open manet.nam w]

\$ns namtrace-all-wireless \$namfd 800 800

Topography

set topo [new Topography]

\$topo load_flatgrid 800 800

Create GOD

create-god 5

Wireless channel

set chan_1_ [new Channel/WirelessChannel]

Node configuration

\$ns node-config \

-adhocRouting AODV \

-llType LL \

-macType Mac/802_11 \

-ifqType Queue/DropTail/PriQueue \

-ifqLen 50 \

-antType Antenna/OmniAntenna \

-propType Propagation/TwoRayGround \

-phyType Phy/WirelessPhy \

-channel \$chan_1_ \

-topoInstance \$topo \

-agentTrace ON \

-routerTrace ON \

-macTrace ON \

-movementTrace ON

Create nodes

for {set i 0} {\$i < 5} {incr i} {

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```
    set node_(i)[ns node]
}
```

```
# Initial positions
```

```
$node_(0) set X_ 100
```

```
$node_(0) set Y_ 100
```

```
$node_(0) set Z_ 0
```

```
$node_(1) set X_ 250
```

```
$node_(1) set Y_ 200
```

```
$node_(1) set Z_ 0
```

```
$node_(2) set X_ 400
```

```
$node_(2) set Y_ 300
```

```
$node_(2) set Z_ 0
```

```
$node_(3) set X_ 550
```

```
$node_(3) set Y_ 400
```

```
$node_(3) set Z_ 0
```

```
$node_(4) set X_ 700
```

```
$node_(4) set Y_ 500
```

```
$node_(4) set Z_ 0
```

```
for {set i 0} {$i < 5} {incr i} {
    $ns initial_node_pos node_$i 20
}
```

```
# UDP Agent
```

```
set udp0 [new Agent/UDP]
```

```
$ns attach-agent $node_(0) $udp0
```

```
# Null Agent
```

```
set null0 [new Agent/Null]
```

```
$ns attach-agent $node_(4) $null0
```

```
# Connect
```

```
$ns connect $udp0 $null0
```

```
# CBR Traffic
```

```
set cbr0 [new Application/Traffic/CBR]
```

```
$cbr0 set packetSize_ 512
```

```
$cbr0 set interval_ 0.1
```

```
$cbr0 attach-agent $udp0
```

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Mobility

nsat1.0"node_(1) setdest 300 300 10"

nsat2.0"node_(2) setdest 450 350 10"

nsat3.0"node_(3) setdest 600 450 10"

Start traffic

nsat1.0"cbr0 start"

nsat8.0"cbr0 stop"

Finish procedure

proc finish {} {

 global ns tracefd namfd

 \$ns flush-trace

 close \$tracefd

 close \$namfd

 exec nam manet.nam &

 exit 0

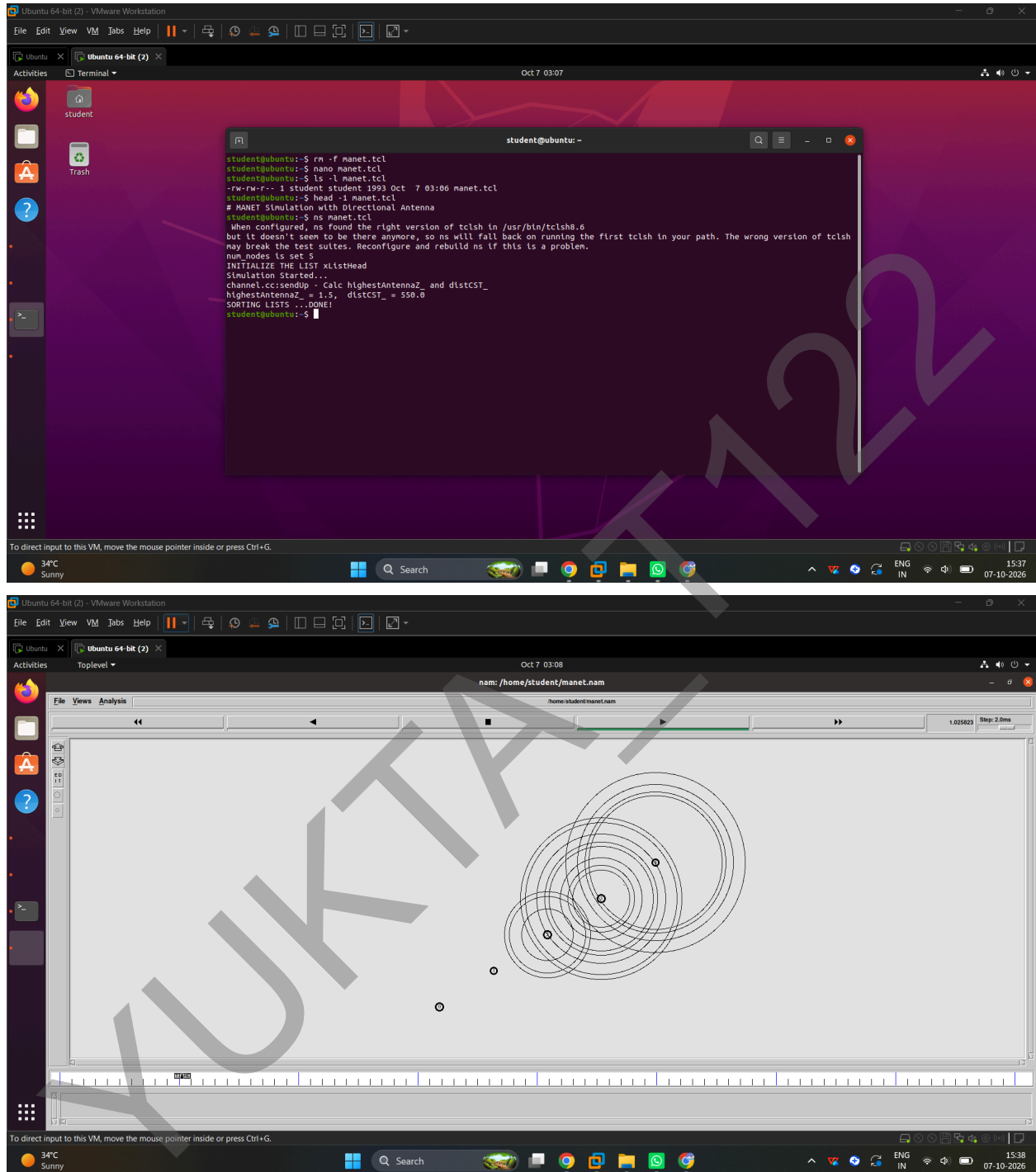
}

\$ns at 10.0 "finish"

puts "Simulation Started..."

\$ns run

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